Math 103 Spring 2021 Midterm Exam Solutions

11 April 2021 20:34

ÖZYEĞİN UNIVERSITY
Math 103/113
Spring 2021

Midterm Exam Solutions

Q1) (Functions) [$6 + 8 = 14$ points]

(a) Given $f(x) = \sqrt{x+2}$ and $g(x) = \dfrac{3}{x}$, write a formula for $f \circ g$ and $g \circ f$, and find the domain and range of each.

(b) Solve for the angle θ, where $0 \le \theta \le 2\pi$.

$$\sin 2\theta + \sqrt{3}\cos\theta = 0$$

(a) $(f \circ g)(x) = f(g(x)) = \sqrt{\frac{3}{x} + 2}$

$\text{Dom}(f \circ g) = (-\infty, \frac{-3}{2}) \cup (0, \infty)$

$\text{Range}(f \circ g) = [0, \sqrt{2}) \cup (\sqrt{2}, \infty)$

$(g \circ f)(x) = g(f(x)) = \dfrac{3}{\sqrt{x+2}}$

$\text{Dom}(g \circ f) = (-2, \infty)$

$\text{Range}(g \circ f) = (0, \infty)$

1 point each

(b) $2\sin\theta\cos\theta + \sqrt{3}\cos\theta = 0$

$\cos\theta(2\sin\theta + \sqrt{3}) = 0$

$\cos\theta = 0, \quad \sin\theta = \dfrac{-\sqrt{3}}{2}$

$$\boxed{\theta = \frac{\pi}{2}, \theta = \frac{3\pi}{2}, \theta = \frac{4\pi}{3}, \theta = \frac{5\pi}{3}}$$

2 points each

Q2) (Limits and Continuity) [12 points]

Given $f(x) = \dfrac{x^2 - 6x + 5}{x^3 - 5x^2}$, find the following limits:

2 points each

$\lim_{x\to 0^-} f(x) = -\infty$ $\quad \lim_{x\to 0^+} f(x) = -\infty$ $\quad \lim_{x\to 0} f(x) = -\infty$

$\lim_{x\to 5^-} f(x) = \frac{4}{25}$ $\quad \lim_{x\to 5^+} f(x) = \frac{4}{25}$ $\quad \lim_{x\to 5} f(x) = \frac{4}{25}$

Since $f(x) = \dfrac{x^2-6x+5}{x^3-5x^2} = \dfrac{(x-5)(x-1)}{x^2(x-5)} = \dfrac{x-1}{x^2}$ for $x \neq 5$.

Q3) (Limits and Continuity) [12 points]

Show that the equation $x^3 - 14x + 2 = 0$ has a solution in the interval $[3, 4]$. **Explain your reasoning in detail.**

$f(x) = x^3 - 14x + 2$ is continuous on $[3,4]$ since it is a polynomial. 4 points

$f(3) = 3^3 - 14(3) + 2 = -13 < 0$

$f(4) = 4^3 - 14(4) + 2 = 10 > 0$

4 points

Since $f(3) < 0 < f(4)$, we have $f(c) = 0$ for some c in $[3,4]$ by the Intermediate Value Theorem. Hence, c is a solution of the given equation. 4 points

Q4) (Derivatives) [12 points]

Find equations of **all** lines that are tangent to the curve $y = \tan x$. $-\pi/2 < x < \pi/2$, which are parallel to the line $y = 2x$.

$$y' = \frac{d}{dx}[\tan x] = \sec^2 x$$

Tangents parallel to $y = 2x$ have slope 2. Hence,

$$y' = \sec^2 x = 2, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$$ 2 points

$$\Rightarrow x = -\frac{\pi}{4} \text{ and } x = \frac{\pi}{4}$$ 4 points

At $x = -\frac{\pi}{4}$, $y = \tan\left(-\frac{\pi}{4}\right) = -1$ and $y' = 2$. So,

$$y - (-1) = 2\left(x - \left(-\frac{\pi}{4}\right)\right) \Rightarrow \boxed{y = 2x + \frac{\pi}{2} - 1}$$ 3 points

At $x = \frac{\pi}{4}$, $y = \tan\left(\frac{\pi}{4}\right) = 1$ and $y' = 2$. So,

$$y - 1 = 2\left(x - \frac{\pi}{4}\right) \Rightarrow \boxed{y = 2x - \frac{\pi}{2} + 1}$$ 3 points

Q5) (Derivatives) [12 points]

Suppose that functions f and g and their derivatives with respect to x have the values shown in the table below.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
2	7	2	1/7	-4
3	1	-8	π	5

Find the following derivatives of the following functions at the given values of x.

(a) $f(x) \cdot g(x)$, $x = 3$

(b) $\dfrac{f(x)}{g(x)}$, $x = 2$

(c) $(f \circ g)(x)$, $x = 2$

(d) $\dfrac{1}{g^2(x)}$, $x = 3$

(a) $\frac{d}{dx}[f(x)g(x)]\Big|_{x=3} = f'(3)g(3) + f(3)g'(3)$

$= (\pi)(-8) + (1)(5)$

$= \boxed{5 - 8\pi}$ 3 points

(b) $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right]\Big|_{x=2} = \dfrac{f'(2)g(2) - f(2)g'(2)}{[g(2)]^2}$

$= \dfrac{(1/7)(2) - (7)(-4)}{(2)^2} = \boxed{\dfrac{99}{14}}$

3 points

(c) $\frac{d}{dx}[f(g(x))]\Big|_{x=2} = f'(g(2)) \cdot g'(2)$

$= f'(2) \cdot g'(2) = \left(\frac{1}{7}\right)(-4) = \boxed{\dfrac{-4}{7}}$

3 points

(d) $\frac{d}{dx}[(g(x))^{-2}]\Big|_{x=3} = -2(g(2))^{-1} g'(2)$

$= -2\left(\dfrac{1}{2}\right)(-4) = \boxed{4}$

3 points

Q6) (Derivatives) [12 points]

Find $\frac{dy}{dx}$ if $x^y = y^x$.

4 points

$$\ln x^y = \ln y^x \Rightarrow y \ln x = x \ln y \Rightarrow \frac{d}{dx}[y \ln x] = \frac{d}{dx}[x \ln y]$$

$$\left(\frac{dy}{dx}\right)(\ln x) + y\left(\frac{1}{x}\right) = (1)(\ln y) + (x)\left(\frac{1}{y} \cdot \frac{dy}{dx}\right)$$

6 points

$$\frac{dy}{dx} = \frac{\ln y - \frac{y}{x}}{\ln x - \frac{x}{y}} = \boxed{\frac{xy \ln y - y^2}{xy \ln x - x^2}}$$

2 points

Q7) (Derivatives) [12 points]

Find a linearization of the function $f(x) = e^{-x}$ at $a = 0$, and use this linearization to approximate $e^{-0.1}$.

$$f(0) = e^{-0} = e^0 = 1$$

4 points

$$f'(x) = \frac{d}{dx}[e^{-x}] = -e^{-x}, \quad f'(0) = -e^{-0} = -1$$

$$L(x) = f(0) + f'(0)(x-0) = 1 + (-1)(x-0)$$

$$= \boxed{1 - x}$$

4 points

$$e^{-0.1} = f(0.1) \approx L(0.1) = 1 - 0 \cdot 1 = \boxed{0.9}$$

4 points

Q8) (Applications of Derivatives) [14 points]

Find the extreme values, and where they occur, of the function $y = x\sqrt{4-x^2}$ on the interval $[-2, 2]$.

$$y' = \frac{d}{dx}\left[x\sqrt{4-x^2}\right] = \frac{4-2x^2}{\sqrt{4-x^2}}$$

4 points

2 points

$y' = 0 \Rightarrow 4-2x^2 = 0 \Rightarrow x^2 = 2 \Rightarrow x = -\sqrt{2},\ x = \sqrt{2}$ are the critical points, and $x=-2$ and $x=2$ are the endpoints:

$$f(-\sqrt{2}) = (-\sqrt{2})\sqrt{4-(-\sqrt{2})^2} = -2$$
$$f(\sqrt{2}) = (\sqrt{2})\sqrt{4-(\sqrt{2})^2} = 2$$
$$f(-2) = (-2)\sqrt{4-(-2)^2} = 0$$
$$f(2) = (2)\sqrt{4-(2)^2} = 0$$

4 points

Since $f(-\sqrt{2}) < f(-2) = f(2) < f(\sqrt{2})$, there is an absolute maximum of 2 at $x=\sqrt{2}$, there is an absolute minimum of -2 at $x=-\sqrt{2}$.

4 points